

PregaCare Project Documentation

1. Project Overview

PregaCare is an AI-powered maternal health support system designed to provide personalized risk assessment and clinical guidance for pregnant women in India. It combines machine learning for clinical risk prediction with regional health infrastructure data (NFHS-5) to provide a holistic safety report.

2. Technology Stack

Backend

- FastAPI: Modern, high-performance web framework for the REST API.
- Python: Primary language for data processing and model inference.
- Uvicorn: ASGI server for running the FastAPI application.
- scikit-learn & XGBoost: Libraries used for the Stacking Ensemble Model.
- Pandas & NumPy: Data manipulation and numerical calculations.

Frontend

- React.js: UI framework for building a responsive user interface.
- Vite: Next-generation frontend tooling for fast development.
- Tailwind CSS: Utility-first CSS framework for premium styling.
- Framer Motion: Animation library for smooth UI transitions.
- Lucide React: Modern icon library.

AI & Data

- OpenRouter API: Gateway used to access Gemini 2.0 Flash Lite.
- RAG (Retrieval-Augmented Generation): System to inject clinical guidelines into AI context.
- NFHS-5 Dataset: Source for regional healthcare infrastructure metrics.

3. Algorithms & Calculations

Clinical Risk Prediction (Stacking Ensemble)

The core clinical risk is predicted using a Stacking Model which ensembles multiple base learners (Random Forest, XGBoost, etc.) to achieve higher accuracy. It processes six primary vitals: Age,

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Systolic BP, Diastolic BP, Blood Sugar, Body Temp, and Heart Rate.

Regional Safety Index (TOPSIS)

The TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) algorithm was used to rank Indian districts based on healthcare infrastructure data from the National Family Health Survey (NFHS-5). This score (0 to 1) represents the relative safety of the patient's location.

Integrated Risk Score Formula

The final 'PregaCare Index' is a weighted combination of clinical health and regional safety:

Combined Score = (Clinical_Risk * 0.75) + ((1 - Safety_Score) * 100 * 0.25)

This weighting prioritizes the patient's immediate clinical vitals (75%) while acknowledging that a lack of local medical infrastructure adds significant risk (25%).

4. Risk Categorization

- Low Risk (0 - 35): Routine care and standard checkups recommended.
- Mid Risk (36 - 70): Moderate concern; increased monitoring and doctor consultation advised.
- High Risk (71 - 100): Immediate medical attention and specialist consultation required.

5. Keywords & Naming Conventions

- 'assessor': Reference to the PregaCareFinalAssessment class.
- 'engine': Reference to the PregaCareAIEngine (RAG system).
- 'vitals': The clinical data object (Age, BP, BS, etc.).
- 'risk_band': The categorical label (Low, Mid, High).
- 'safety_index': The normalized TOPSIS score for a district.